

單選題，共 25 題，每題 4 分（答錯不倒扣）

1. Which of the following is true for binary search algorithm?
 - (A) Can be used on unsorted data
 - (B) Time complexity is $O(n)$
 - (C) Data must be sorted
 - (D) Always returns the first occurrence
 - (E) Requires linear time
2. Which logic gate has the following truth table: output is 1 only if inputs are different?
 - (A) AND
 - (B) OR
 - (C) NAND
 - (D) XOR
 - (E) XNOR
3. In an operating-system context, the banker's algorithm is used to:
 - (A) Detect deadlock
 - (B) Prevent starvation
 - (C) Avoid deadlock by resource allocation
 - (D) Select victim pages for replacement
 - (E) Schedule disk I/O requests
4. The IP, TCP, and UDP headers all carry a 16-bit field named "checksum." Which ISO/OSI layer primarily uses that field?
 - (A) Transport
 - (B) Data-link
 - (C) Network
 - (D) Physical
 - (E) None of the above
5. In IPv4, a /26 CIDR network supports how many usable host addresses?
 - (A) 30
 - (B) 62
 - (C) 64
 - (D) 32
 - (E) 26

6. IPv6 eliminates the need for which IPv4 feature?
- (A) DHCP
 - (B) ARP
 - (C) DNS
 - (D) ICMP
 - (E) NAT
7. Message Authentication Codes (MACs) provide:
- (A) Integrity and authenticity
 - (B) Non-repudiation and confidentiality
 - (C) Confidentiality only
 - (D) Key exchange
 - (E) Forward secrecy
8. A max-heap stored in an array of length n guarantees that the parent of the node at index i is at:
- (A) $\text{floor}(i / 2)$
 - (B) $\text{floor}((i - 1)/2)$
 - (C) $\text{ceil}(i / 2)$
 - (D) $i - 1$
 - (E) $\text{floor}(i / 3)$
9. In 8-bit two's-complement arithmetic, adding $0x73 + 0x94$ raises the overflow flag because:
- (A) Signs differ but result is negative
 - (B) Both operands are positive but result is negative
 - (C) Carry-in to MSB = carry-out from MSB
 - (D) Parity bit = 0
 - (E) Half-carry occurred

10. A 64-MiB main memory is byte-addressable. A 4-way set-associative cache of 32 KiB uses 64-byte lines. Number of tag bits in a 32-bit address is:
- (A) 19
 - (B) 16
 - (C) 13
 - (D) 20
 - (E) 21
11. In standard C, whether char is signed or unsigned by default is:
- (A) Always signed
 - (B) Always unsigned
 - (C) Controlled by pragma
 - (D) Undefined behaviour
 - (E) Implementation-defined
12. You are designing an ALU that must support the rotate-through-carry left (RCL) operation on 32-bit registers. If the current CF = 1 and the operand is 0x80000000, what are the new value of the register and the new carry flag?
- (A) 0x00000000, CF=0
 - (B) 0x00000000, CF=1
 - (C) 0x00000001, CF=0
 - (D) 0x00000001, CF=1
 - (E) 0x00000003, CF=1
13. You benchmark QuickSort and Measure that its running time on random permutations doubles when n grows from $2M$ to $4M$. Which average-case complexity class is most consistent with this observation?
- (A) $O(n)$
 - (B) $O(n \log \log n)$
 - (C) $O(n \log n)$
 - (D) $O(\sqrt{n})$
 - (E) $O(n^2)$

14. A university has the prefix 2001:0db8:acad::/48 and wants 400 subnets. What is the smallest subnet mask that meets the need and keeps addresses aligned on nibble boundaries?
- (A) /52
 - (B) /56
 - (C) /48
 - (D) /60
 - (E) /64
15. A 900 MHz microcontroller must process each PWM cycle in ≤ 90 instruction cycles. An ISR currently compiles to 112 cycles. Which optimisation meets timing with least code change?
- A) Unroll loops
 - B) Inline all function calls
 - C) Move ISR to lower priority
 - D) Switch to Thumb-2
 - E) None of the above
16. A function returns `std::vector<T> foo()`; . At call-site you write:

```
std::vector<T> v;  
v = std::move(foo());
```

Why might the extra `std::move()` be worse than writing `v = foo();`?

- A) Disables copy elision
- B) Prevents RVO
- C) Converts to lvalue reference
- D) Increases alignment padding
- E) None of the above

17. A company's e-commerce database has a "products" table with 10 million rows. A query "SELECT * FROM products WHERE price BETWEEN 100 AND 200" takes 45 seconds. What optimization would provide the most significant performance improvement?
- (A) Add more RAM to the database server
 - (B) Increase the number of CPU cores
 - (C) Partition the table by product category
 - (D) Use a faster SSD storage
 - (E) Create an index on the price column
18. A processor has a 32KB L1 cache with 64-byte lines. A program accesses a 2D array of 1024×1024 integers (4 bytes each) in column-major order. Assuming the cache is initially empty, approximately how many cache misses occur in the first complete column access?
- (A) 1024
 - (B) 4096
 - (C) 16384
 - (D) 65536
 - (E) 1048576
19. A company implements AES-256 encryption for sensitive data. Each encryption operation takes 0.1ms. If they need to encrypt 100GB of data in 1-hour batches, what is the minimum number of encryption threads needed to meet the deadline?
- (A) 100
 - (B) 200
 - (C) 400
 - (D) 1000
 - (E) 2778

20. A real-time system has 3 periodic tasks: T1 (period=50ms, execution=10ms), T2 (period=100ms, execution=20ms), T3 (period=200ms, execution=40ms). Using Rate Monotonic Scheduling, what is the CPU utilization?

- (A) 60%
- (B) 70%
- (C) 80%
- (D) 90%
- (E) 100%

21. What is the output of the following C++ code?

```
#include <iostream>
using namespace std;

class Test {
public:
    Test() { cout << "Constructor "; }
    ~Test() { cout << "Destructor "; }
    Test(const Test& t) { cout << "Copy "; }
};

int main() {
    Test t1;
    Test t2 = t1;
    Test t3(t1);
    return 0;
}
```

- (A) Constructor Copy Constructor Destructor Destructor Destructor
- (B) Constructor Constructor Copy Destructor Destructor Destructor
- (C) Constructor Copy Copy Destructor Destructor
- (D) Constructor Copy Copy Destructor Destructor Destructor
- (E) Constructor Constructor Constructor Destructor Destructor Destructor

22. What is the output of the following code?

```
#include <iostream>
using namespace std;

void function(int x) {
    cout << "int: " << x << " ";
}

void function(double x) {
    cout << "double: " << x << " ";
}

void function(int x, int y = 10) {
    cout << "two ints: " << x << ", " << y << " ";
}

int main() {
    function(5);
    function(5.0);
    function(5, 15);
    return 0;
}
```

- (A) int: 5 double: 5 two ints: 5,15
- (B) int: 5 double: 5.0 two ints: 5,15
- (C) two ints: 5,10 double: 5.0 two ints: 5,15
- (D) Run time error due incompatible type
- (E) Compilation error due to ambiguous function call

23. What is the output of the following code?

```
#include <iostream>
using namespace std;

void function(const int& x) {
    cout << "const ref: " << x << " ";
}

void function(int& x) {
    cout << "ref: " << x << " ";
    x = 100;
}

int main() {
    int a = 5;
    const int b = 10;

    function(a);
    function(b);
    cout << a << endl;

    return 0;
}
```

- (A) const ref: 5 const ref: 10 5
- (B) ref: 5 ref: 10 100
- (C) ref: 5 const ref: 10 100
- (D) const ref: 5 ref: 10 5
- (E) Compilation error

24. What is the output of the following code?

```
#include <iostream>

int main() {
    int count = 0;
    for (int i = 0; i < 3; ++i) {
        for (int j = 0; j < 3; ++j) {
            if (i == j) {
                continue;
            }
            if ((i + j) % 2 == 0) {
                count++;
            }
        }
    }
    std::cout << count << std::endl;
    return 0;
}
```

- (A) 0
- (B) 2
- (C) 3
- (D) 4
- (E) 6

25. What is the output of the following code?

```
#include <iostream>

int totalised(int n)
{
    static int accum = 0;
    if (n == 0) return accum;
    accum += n;
    return n + totalised(n - 1);
}

int main()
{
    std::cout << totalised(3) << ' '
               << totalised(2) << '\n';
}
```

- (A) 2 3
- (B) 6 9
- (C) 6 6
- (D) 12 10
- (E) 12 12